

25.5 Bad controls: mediators and colliders

A regression that “controls for” many variables is not better for it. A control helps when it blocks a back-door path, costs at most precision when it lies on no path, and does harm when it lies on the causal path or when conditioning on it opens a closed path. Angrist and Pischke (2009) call the last two kinds **bad controls**. This section computes the damage in the linear models of Figure 25.5.1 and describes the related bias from selecting the sample on a collider.

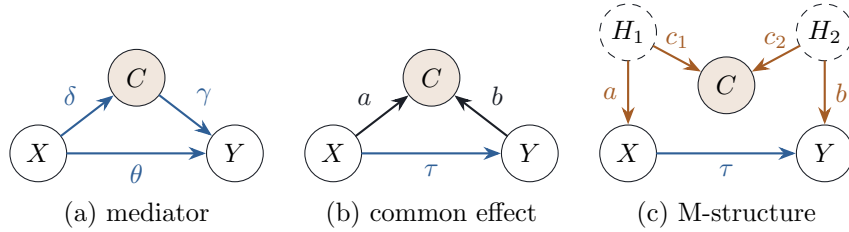


Figure 25.5.1. Three bad controls C (shaded). (a) C is a mediator. (b) C is a common effect of X and Y . (c) C is measured before X but is a collider on the back-door path through the unrecorded H_1 and H_2 .

25.5.1 Three computations

Theorem 25.5.1 — Bad controls in linear models

In each model below the disturbances U_X, U_Y, U_C (and, in (c), the unrecorded variables H_1, H_2) are independent with mean zero and variances $\sigma_X^2, \sigma_Y^2, \sigma_C^2 > 0$ (and 1). Let τ be the total effect of X on Y . In every case the slope of $L(Y | X)$ equals τ , so the simple regression is consistent, while the coefficient $\beta_{X|C}$ of X in $L(Y | X, C)$ is as follows.

- (a) **Mediator:** $X = U_X$, $C = \delta X + U_C$, $Y = \theta X + \gamma C + U_Y$. Here $\tau = \theta + \gamma\delta$ and $\beta_{X|C} = \theta$: adjustment removes the indirect effect $\gamma\delta$.
- (b) **Common effect:** $X = U_X$, $Y = \tau X + U_Y$, $C = aX + bY + U_C$. Then

$$\beta_{X|C} = \tau - \frac{(a + b\tau) b \sigma_Y^2}{b^2 \sigma_Y^2 + \sigma_C^2}. \quad (25.5.1)$$

- (c) **M-structure:** $X = aH_1 + U_X$, $C = c_1H_1 + c_2H_2 + U_C$, $Y = \tau X + bH_2 + U_Y$. Then

$$\beta_{X|C} = \tau - \frac{a b c_1 c_2}{D}, \quad (25.5.2)$$

$$D = (a^2 + \sigma_X^2)(c_1^2 + c_2^2 + \sigma_C^2) - a^2 c_1^2 > 0.$$

Proof.

- (a) $Y - \theta X - \gamma C = U_Y$ is uncorrelated with X and C , so $L(Y | X, C) = \theta X + \gamma C$. Without C , $Y = (\theta + \gamma\delta)X + (\gamma U_C + U_Y)$, and the bracket is uncorrelated with X , so the slope is τ .
- (b) Without C , U_Y is uncorrelated with X and the slope is τ . With C , write $E =$

$C - (a + b\tau)X = bU_Y + U_C$, which is uncorrelated with X . The span of $1, X, C$ is the span of $1, X, E$, with E orthogonal to 1 and X , so $L(U_Y | X, C) = L(U_Y | X) + L(U_Y | E) = 0 + \lambda E$, where $\lambda = \text{Cov}(U_Y, E) / \text{Var}(E) = b\sigma_Y^2 / (b^2\sigma_Y^2 + \sigma_C^2)$. Hence $L(Y | X, C) = \tau X + \lambda C - \lambda(a + b\tau)X$, whose coefficient of X is [Eq. 25.5.1](#).

- (c) Without C , $bH_2 + U_Y$ is uncorrelated with X and the slope is τ . With C , $L(Y | X, C) = \tau X + bL(H_2 | X, C)$, since U_Y is uncorrelated with X and C . The coefficients of $L(H_2 | X, C)$ solve the population equations of Theorem 6.11.1, with covariance matrix of (X, C) equal to

$$\begin{pmatrix} a^2 + \sigma_X^2 & ac_1 \\ ac_1 & c_1^2 + c_2^2 + \sigma_C^2 \end{pmatrix}$$

and right-hand side $(\text{Cov}(X, H_2), \text{Cov}(C, H_2)) = (0, c_2)$. The determinant is D , positive since the matrix is a positive definite covariance matrix, and the first coefficient is $-ac_1c_2/D$. Multiply by b .

■

In (a) the adjusted coefficient is the **direct effect**, with the mediator held fixed by intervention. That may be of interest, but only if nothing confounds the mediator and the outcome ([Exercise \(B2\)](#)); a policy decision needs τ ([Section 18.1](#) gave the experimental version).

In (b) the bias exists even when $\tau = 0$: adjusting for a consequence of both variables produces an “effect” $-ab\sigma_Y^2 / (b^2\sigma_Y^2 + \sigma_C^2)$ where there is none, negative when a and b have the same sign. This is the collider of Proposition 25.3.3(c).

In (c) the control is measured before X , so the advice to adjust only for pre-treatment variables does not exclude it, yet adjustment opens the back-door path $X \leftarrow H_1 \rightarrow C \leftarrow H_2 \rightarrow Y$. This **M-bias**, named after the graph’s shape, was pointed out by Greenland, Pearl and Robins (1999). It is a product of four path coefficients over D and tends to be small unless all four are large (Ding and Miratrix 2015). When the collider is also a confounder, with arrows $C \rightarrow X$ and $C \rightarrow Y$, neither choice removes the bias; Ding and Miratrix call this butterfly bias.

Example. (Three bad controls by simulation)

Take, with unit variances throughout:

- a mediator with $\theta = 0.5$, $\delta = 1.2$, $\gamma = 1$, so that $\tau = 1.7$ and $\beta_{X|C} = 0.5$;
- a common effect with $\tau = 1$, $a = 0.5$, $b = 1$, for which [Eq. 25.5.1](#) gives $\beta_{X|C} = 1 - (1.5)(1)/2 = 0.25$;
- an M-structure with $\tau = 1$ and $a = b = c_1 = c_2 = 1.5$, for which $D = \frac{13}{4} \cdot \frac{11}{2} - \frac{81}{16} = \frac{205}{16} = 12.8125$ and $\beta_{X|C} = 1 - \frac{81}{205} = 0.605$.

Over 4000 samples of $n = 400$, the least squares coefficients of X without and with C average 1.700 and 0.499 for the mediator, 1.000 and 0.251 for the common effect, and 0.999 and 0.605 for the M-structure ([Figure 25.5.2](#)). In the M-structure the adjusted estimate is also slightly *less* variable (standard deviation 0.046 against 0.050), a reminder that a smaller standard error says nothing about bias.

```
import numpy as np

rng = np.random.default_rng(2505)
```

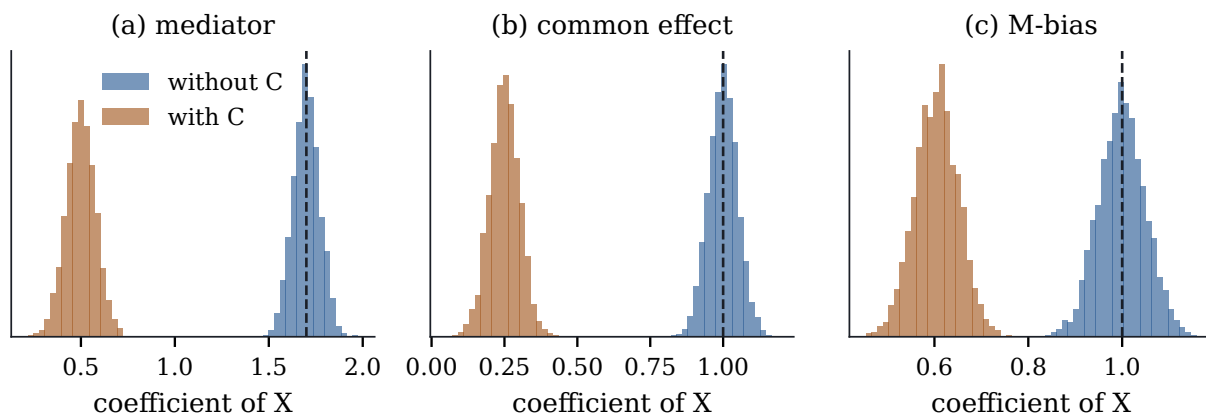


Figure 25.5.1: Sampling distributions of the least squares coefficient of X without (blue) and with (brown) adjustment for C , in the three models of [Example \(Three bad controls by simulation\)](#); 4000 samples of size 400. The dashed line is the total effect τ .

```
def mediator(n):          #  $X \rightarrow C \rightarrow Y$  and  $X \rightarrow Y$ ; total effect  $0.5 + 1.2 * 1.0 = 1.7$ 
    x = rng.normal(size=n); c = 1.2 * x + rng.normal(size=n)
    return x, c, 0.5 * x + 1.0 * c + rng.normal(size=n)

def collider(n):          #  $X \rightarrow Y$ , and  $X \rightarrow C \leftarrow Y$ ; total effect 1
    x = rng.normal(size=n); y = 1.0 * x + rng.normal(size=n)
    return x, 0.5 * x + 1.0 * y + rng.normal(size=n), y

def m_bias(n):            #  $X \leftarrow H1 \rightarrow C \leftarrow H2 \rightarrow Y$ , and  $X \rightarrow Y$ ; total effect 1
    h1, h2 = rng.normal(size=(2, n))
    x = 1.5 * h1 + rng.normal(size=n)
    c = 1.5 * h1 + 1.5 * h2 + rng.normal(size=n)
    return x, c, 1.0 * x + 1.5 * h2 + rng.normal(size=n)

def slopes(x, c, y):
    """Coefficient of  $x$  without and with adjustment for  $c$ ."""
    one = np.ones(len(x))
    b_short = np.linalg.lstsq(np.column_stack([one, x]), y, rcond=None)[0][1]
    b_long = np.linalg.lstsq(np.column_stack([one, x, c]), y, rcond=None)[0][1]
    return b_short, b_long
```

[Run this code online](#)

```
n, reps = 400, 500
for name, model in [("mediator", mediator), ("collider", collider), ("M-bias",
    m_bias)]:
    est = np.array([slopes(*model(n)) for _ in range(reps)])
    print(f"{name:9s} mean without C {est[:, 0].mean():.3f}    mean with C {est[:,
    1].mean():.3f}")
```

[Run this code online](#)

25.5.2 Post-treatment variables in general

A variable affected by X removes part of the effect if it lies on a directed path to Y , and opens a path if Y or a cause of Y also affects it; its descendants act partially (Exercise (B1)). A descendant of X with no other connection to Y costs only precision (Exercise (B1)). “Adjust only for pre-treatment variables” is sound in randomized experiments (Section 25.6); in observational data it is neither sufficient, by the M-structure, nor necessary, since a later measurement can proxy an unrecorded confounder. The graph, not the timing, decides. Adjusting for a pure cause of X can even *increase* the bias from an unrecorded confounder (Exercise (B3)); Pearl (2010) calls such variables bias amplifiers.

25.5.3 Selection on a collider

A sample selected on a common effect of X and Y is conditioned on a collider without any adjustment. Berkson (1946) saw two diseases, independent in the population, associated among hospital patients, since either brings a patient to hospital; Elwert and Winship (2014) survey this *endogenous selection bias*. In the common-effect model of Example (Three bad controls by simulation), observing only the units with $C > 1$ (about 0.31 of them) moves the slope of Y on X from 1.002 to 0.584.

```
x, c, y = collider(200_000)
keep = c > 1                                # only units with a large C
are observed
slope_all = np.polyfit(x, y, 1)[0]
slope_sel = np.polyfit(x[keep], y[keep], 1)[0]
print(f"slope of Y on X: all units {slope_all:.3f}, units with C > 1 {slope_sel:.3f}")
```

[Run this code online](#)

Warning. A variable belongs in a causal regression because of its position in the graph, not because it improves R^2 or is “significant”; predictive model selection (Chapter 29) readily selects mediators and colliders.

25.5.4 Exercises

25.5.4.1 A. Check your understanding

Exercise A1.

A study of the effect of a training programme (X) on earnings two years later (Y) considers adjusting for: (i) education before the programme; (ii) the job obtained six months after the programme; (iii) whether the person was still living in the region at the end of the study, which depends on both training and earnings; (iv) the day of the week on which the person applied. Classify each as a confounder, mediator, collider or irrelevant, under a plausible graph, and say which should be adjusted for.

25.5.4.2 B. Practice

Exercise B1.

Let $X = U_X$, $Y = \tau X + U_Y$ and $D = dX + U_D$, where D is affected by X but has no connection with Y . Show that the coefficient of X in $L(Y | X, D)$ is τ , and that the asymptotic variance of the least squares coefficient is larger with D than without it by the factor $(d^2\sigma_X^2 + \sigma_D^2)/\sigma_D^2$.

Exercise B2.

In the mediator model add an unrecorded U , with variance σ_U^2 , that affects both the mediator and the outcome: $C = \delta X + gU + U_C$, $Y = \theta X + \gamma C + hU + U_Y$. Show that

$$\beta_{X|C} = \theta - \frac{\delta g h \sigma_U^2}{g^2 \sigma_U^2 + \sigma_C^2},$$

so that adjusting for the mediator does not even recover the direct effect θ .

Solution. $Y = \theta X + \gamma C + hU + U_Y$ and U_Y is uncorrelated with (X, C) , so $\beta_{X|C} = \theta + h\kappa$, where κ is the coefficient of X in $L(U | X, C)$. Put $E = C - \delta X = gU + U_C$, uncorrelated with X . As in the proof of [Theorem 25.5.1\(b\)](#), $L(U | X, C) = L(U | E) = \lambda(C - \delta X)$ with $\lambda = g\sigma_U^2/(g^2\sigma_U^2 + \sigma_C^2)$, so $\kappa = -\lambda\delta$. The mediator is a collider on the path $X \rightarrow C \leftarrow U \rightarrow Y$, which adjustment opens.

Exercise B3.

Let U be unrecorded, $X = \pi I + gU + U_X$ and $Y = \tau X + hU + U_Y$, with I, U, U_X, U_Y independent, $\text{Var } I = \text{Var } U = 1$ and $\text{Var } U_X = \sigma^2$. Show that the slope of $L(Y | X)$ is $\tau + gh/(\pi^2 + g^2 + \sigma^2)$, and that the coefficient of X in $L(Y | X, I)$ is $\tau + gh/(g^2 + \sigma^2)$. Adjusting for the instrument I enlarges the bias.

Solution. $\text{Cov}(X, Y) = \tau \text{Var } X + gh$ and $\text{Var } X = \pi^2 + g^2 + \sigma^2$, which gives the first slope. For the second, by Proposition 14.5.2 the coefficient is $\sigma_{YX \cdot I}/\sigma_{XX \cdot I}$. Removing the linear effect of I leaves $X - \pi I = gU + U_X$, with variance $g^2 + \sigma^2$, and $Y - L(Y | I) = \tau(gU + U_X) + hU + U_Y$, so $\sigma_{YX \cdot I} = \tau(g^2 + \sigma^2) + gh$. Divide. The confounded covariance gh is the same in both, but adjustment divides it by a smaller variance.